

Re-investigate Word Concreteness Effect Using Z-Curve Analysis

Ruifeng Yu, Jie Yee Lim, Chenghan Xie, Debbie Hsu, Yuexi Zhang

PSYC 6018: Research Design

Dr. Rick Thomas

May 4, 2026

Abstract

The word concreteness effect, the finding that concrete words are processed faster than abstract words, is one of the most well-known effects in psycholinguistics. However, the credibility of this literature has not been systematically evaluated using modern bias-detection methods. The present study applied Z-curve 2.0 analysis to 45 statistically significant focal tests extracted from published studies on the word concreteness effect. Results revealed a substantial discrepancy between the observed discovery rate (ODR = 1.00) and the expected discovery rate (EDR = .15, 95% CI [.05, .75]), suggesting that the published record likely overrepresents significant findings relative to what the underlying statistical power would predict. The expected replicability rate was .63 (95% CI [.42, .82]), and the expected false discovery rate was .29 (95% CI [.02, 1.00]). These findings indicate that while the concreteness effect likely reflects a genuine phenomenon, its apparent robustness is inflated by publication bias and selective reporting. Implications for open science practices in psycholinguistics research are discussed.

Keywords: word concreteness effect, Z-curve analysis, publication bias, replicability

Re-investigate Word Concreteness Effect Using Z-Curve Analysis

The word concreteness effect refers to the effect that words representing concrete objects (e.g., table and stone) are recognized faster than words representing abstract objects (e.g., justice and beauty) in visual word recognition tasks (Paivio & Begg, 1971; Jessen et al., 2000). This effect is theoretically important because it informs how semantic meaning is processed, how our mental lexicon is organized, and how semantic knowledge is represented. However, the credibility of the published evidence supporting this effect has not been rigorously evaluated.

A major concern in psychological research is that published findings may overrepresent statistically significant results (Sterling, 1959). Publication bias occurs when studies with significant or theory-consistent findings are more likely to be published than studies with null or inconsistent results (Rosenthal, 1979). Relatedly, questionable research practices (QRPs), such as selectively reporting dependent variables, excluding participants after inspecting results, trying multiple analytic specifications, or emphasizing only significant comparisons, can make published findings appear stronger and more consistent than they truly are (John et al., 2012). These practices are especially consequential when individual studies have modest sample sizes or when many alternative analyses are possible. In such cases, literature may contain many significant findings even if the true effects are smaller, less stable, or less replicable than the published record suggests (Ioannidis, 2005).

Replication is therefore essential for evaluating whether an effect is robust. A statistically significant result in a single study does not necessarily imply that the effect would reappear in a direct replication, especially when sample sizes are modest or analytic flexibility is high. For an

established effect such as word concreteness, the key question is not only whether the effect exists, but whether the published evidence accurately reflects its reliability.

Z-curve 2.0 analysis provides a useful method for addressing this question (Bartoš & Schimmack, 2022). Z-curve estimates the expected discovery rate (EDR) and expected replicability rate (ERR) from the distribution of statistically significant test statistics in the literature. Rather than assuming that all significant findings provide equally strong evidence, Z-curve examines whether the pattern of *z*-scores is consistent with a literature containing high-powered and replicable effects or with a literature shaped by selective reporting and publication bias. This makes Z-curve particularly appropriate for evaluating the word concreteness effect, because the field contains many statistically significant behavioral findings but may also involve heterogeneous tasks, measures, and analytic decisions. By converting focal statistical tests into a common *z*-score metric, Z-curve allows us to evaluate the evidential structure of the literature across different study designs.

The present study re-investigates the credibility of the word concreteness effect using Z-curve analysis. Specifically, we ask three questions. First, does the observed discovery rate (ODR) in the published literature exceed the expected discovery rate estimated from the distribution of significant results? Second, what is the expected replicability rate of published significant findings on the word concreteness effect? Third, do the Z-curve estimates suggest that the literature reflects a robust and replicable effect, or a pattern inflated by publication bias and selective reporting? By addressing these questions, the present analysis aims to provide a more cautious and cumulative assessment of the evidential strength behind the word concreteness effect.

Methods

Preregistration Summary

This study was preregistered on AsPredicted (ID: 275830), prior to any data collection. The preregistered hypotheses, analysis plans, and inclusion criteria are described in the relevant sections below. One deviation from the preregistered plan was made: the publication year eligibility criterion was revised from studies published after 2000 to studies published from 1980 onward, as the original cutoff yielded an insufficient number of eligible publications.

Literature Search and Inclusion Criteria

Studies were identified through a search of ten target journals, including *Cognition*, *Journal of Memory and Language*, *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *Journal of Experimental Psychology: Human Perception and Performance*, *Psychological Science*, *Brain and Language*, *Cortex*, *NeuroImage*, *Journal of Cognitive Neuroscience*, and *Psychophysiology*. Studies were included if they (a) reported empirical research with quantitative behavioral data (e.g., reaction times, accuracy rates); (b) involved healthy human participants without reported neurological, psychiatric, or language-related diagnoses; (c) directly manipulated or compared word concreteness with appropriate statistical controls; (d) employed word recognition tasks such as lexical decision, word naming, semantic categorization, or word identification; (e) reported sufficient statistical information to extract or calculate effect sizes; and (f) were published in or after 1980. Reviews, meta-analyses, theoretical papers, commentaries, and studies examining concreteness only in post-hoc or exploratory analyses were excluded. A total of 45 papers meeting all inclusion criteria were extracted for analysis.

Data Extraction, Focal Tests, and Double-Coding Reliability

A focal test was defined as a primary statistical test directly comparing concrete and abstract word processing. When a study reported multiple sub-comparisons, coders extracted the test most directly relevant to the main concreteness effect. The project leader assigned each of the 45 eligible tests to a pair of coders, with assignments tracked in a shared extraction template. Each coder extracted data independently before the pair compared entries. The initial agreement rate was 84% (38 of 45 tests coded identically). The seven discrepancies arose from (a) ambiguity about which test to extract when studies reported multiple sub-comparisons, and (b) studies reporting both standardized and unstandardized coefficients. Ambiguous cases were resolved through team voting during group meetings. For studies reporting both standardized and unstandardized coefficients, coders were instructed to extract standardized coefficients when available, defaulting to unstandardized when no standardized estimate was reported or when it was insufficient to calculate a p -value.

Z-Curve Analytic Approach

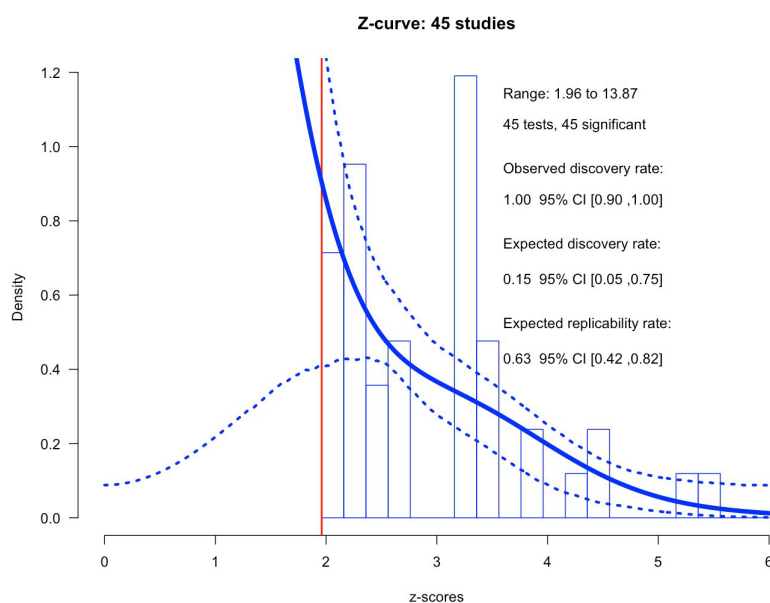
Reported test statistics (t , F , r , χ^2 , or z) were converted to two-tailed p -values in R and transformed to absolute z -scores. Only statistically significant z -scores ($|z| \geq 1.96$) were submitted to the z -curve model, fitted using the *zcurve* package in R (Bartoš & Schimmack, 2022). The model estimates the ODR, EDR, ERR, and EFDR. Evidence of selection bias was evaluated by testing whether 95% confidence intervals of the EDR and ODR overlapped, with non-overlap indicating inflation of significant results beyond what the underlying power would predict. The credibility of the significant findings was assessed by comparing the EFDR against the 5% benchmark proposed by Soric (1989).

Results

Figure 1 shows the results of a z-curve analysis (Bartoš & Schimmack, 2022) conducted on all 45 significant test statistics included in the review. The figure displays the distribution of observed z-scores along with the fitted z-curve and 95% confidence intervals. The ODR was 1.00, 95% CI [.90, 1.00], indicating that every test in the sample was statistically significant. The EDR was substantially lower at .15, 95% CI [.05, .75], suggesting that only about 15% of tests would be expected to reach significance given the estimated mean power of these studies. The expected false discovery rate (EFDR; Sorić, 1989), derived from the EDR, was .29, 95% CI [.02, 1.00], indicating that approximately 29% of the significant findings in this literature may be false positives. The ERR, which reflects the estimated mean power of the significant tests, was .63, 95% CI [.42, .82], indicating that roughly 63% of these results would be expected to reach significance again if the studies were directly replicated.

Figure 1.

Z-Curve Analysis of Word Concreteness Effect Studies ($k = 45$)



Discussion

The present Z-curve analysis reveals a pronounced divergence between nominal statistical significance and the underlying evidential structure of the literature on the word concreteness effect. Although all extracted focal tests were statistically significant ($ODR = 1.00$), the expected discovery rate was substantially lower ($EDR = .15$), suggesting that the published record overrepresents statistically significant findings relative to what would be expected given the estimated distribution of statistical power.

This pattern highlights a central limitation of relying on statistical significance as a proxy for evidential strength. A body of exclusively significant findings may be interpreted as strong support for a theoretical effect; however, Z-curve results demonstrate that such uniformity can also emerge from selective reporting processes, including publication bias and analytic flexibility. In the present case, the large gap between observed and expected discovery rates is consistent with literature in which non-significant or weaker findings are underreported, thereby inflating the apparent consistency of the effect (Rosenthal, 1979).

At the same time, the distribution of observed scores suggests that the concreteness effect is not solely an artifact of bias. The presence of a wide range of z -values, including several well above conventional significance thresholds, indicates that the literature likely contains genuine signals. This interpretation is further supported by the expected replicability rate ($ERR = .63$), which suggests that approximately two-thirds of the significant findings would be expected to replicate under identical conditions. Thus, while the effect appears to be real, its apparent robustness is overstated when evaluated solely through the lens of published statistically significant results. Conclusions about the strength of the effect should be grounded in cumulative evidence rather than isolated significant results.

The interpretation of these findings depends on several analytic choices and assumptions. First, the Z-curve model relies exclusively on statistically significant results, which may amplify discrepancies between observed and expected discovery rates. Second, the aggregation of heterogeneous statistical tests into a common z-score metric introduces potential estimation error, particularly when underlying study designs and measurement approaches vary. Third, potential violations of independence across studies, such as overlapping samples or shared stimuli, may further inflate the apparent consistency of results. Finally, the relatively small number of focal tests ($k = 45$) increases sensitivity to extreme values and may limit the precision of parameter estimates.

In addition, several limitations of the Z-curve framework should be acknowledged. Because the method does not incorporate non-significant findings, it cannot fully reconstruct the complete evidential landscape of a research domain. In addition, its estimates depend on assumptions about the distribution of true effect sizes and do not identify the specific mechanisms (e.g., publication bias vs. p-hacking) that generate selection effects. Furthermore, Z-curve focuses on statistical significance rather than effect size magnitude, limiting its ability to inform conclusions about the practical importance of observed effects.

Future research should extend the present analysis through additional robustness checks and complementary methods. Sensitivity analyses excluding extreme z-values, subgroup analyses by task type or methodological characteristics, and comparisons with alternative bias-detection approaches (e.g., p-curve, selection models) would provide converging evidence. Incorporating unpublished or non-significant findings would further improve the accuracy of inferences about the evidential structure of the literature.

The present findings suggest that the word concreteness effect is supported by genuine empirical signal but is embedded within a literature that overstates its reliability. These results underscore the importance of moving beyond statistical significance as a primary criterion for scientific credibility and highlight the value of meta-scientific tools such as Z-curve for evaluating the robustness of established effects.

Conclusion

The present study applied Z-curve 2.0 analysis to evaluate the evidential strength of the word concreteness effect in published research. The results indicate that while the effect is likely real, the published literature substantially overstates its reliability. The observed discovery rate of 1.00 stands in stark contrast to the estimated expected discovery rate of .15, pointing to meaningful inflation driven by publication bias and selective reporting. The expected replicability rate of .63 suggests that roughly one-third of published significant findings would fail to replicate under identical conditions, suggesting that statistical significance alone is an insufficient standard for scientific credibility.

These findings carry broader implications for open science practice. First, researchers and reviewers should resist treating a pattern of uniformly significant results as evidence of a robust effect; such uniformity may itself be a diagnostic signal of selective reporting. Second, pre-registration and full reporting of null results are essential for constructing a more accurate picture of effect reliability. Third, cumulative meta-scientific tools such as Z-curve should be routinely applied when evaluating established effects, particularly in fields where analytic flexibility and modest sample sizes are common.

References

- Bartoš, F., & Schimmack, U. (2022). Z-curve 2.0: Estimating replication and discovery rates. *Meta-Psychology*, 6. <https://doi.org/10.15626/MP.2021.2720>
- Ioannidis, J. P. A. (2005). Why most published research findings are false. *PLoS Medicine*, 2(8), e124. <https://doi.org/10.1371/journal.pmed.0020124>
- Jessen, F., Heun, R., Erb, M., Granath, D. O., Klose, U., Papassotiropoulos, A., & Grodd, W. (2000). The concreteness effect: Evidence for dual coding and context availability. *Brain and language*, 74(1), 103-112.
- John, L. K., Loewenstein, G., & Prelec, D. (2012). Measuring the prevalence of questionable research practices with incentives for truth telling. *Psychological Science*, 23(5), 524–532. <https://doi.org/10.1177/0956797611430953>
- Paivio, A., & Begg, I. (1971). Imagery and comprehension latencies as a function of sentence concreteness and structure. *Perception & Psychophysics*, 10(6), 408-412.
- Rosenthal, R. (1979). The file drawer problem and tolerance for null results. *Psychological Bulletin*, 86(3), 638–641.
- Sorić, B. (1989). Statistical "discoveries" and effect-size estimation. *Journal of the American Statistical Association*, 84(406), 608–610. <https://doi.org/10.1080/01621459.1989.10478811>

Sterling, T. D. (1959). Publication decisions and their possible effects on inferences drawn from tests of significance—or vice versa. *Journal of the American Statistical Association*, 54(285), 30–34. <https://doi.org/10.2307/2282137>